

Ritvik Mahajan

rm17.work | +91-8595997599 | ritvikmahajan2001@gmail.com

EDUCATION

THAPAR UNIVERSITY

B.E IN COMPUTER SCIENCE

2019 - 2023

CGPA: 8.13 / 10

DELHI PUBLIC SCHOOL

AISSCE (CLASS XII)

2019 | Vasant Kunj, New Delhi

Aggregate: 92%

NAVY CHILDREN SCHOOL

AISSE (CLASS X)

2017 | New Delhi

CGPA: 10 / 10

LINKS

Leetcode:// [ritvikmahajan17](#)

Github:// [ritvikmahajan17](#)

LinkedIn:// [ritvikmahajan17](#)

Codechef:// [ritvik17](#)

SKILLS

PROGRAMMING

Languages:

- Python • Typescript • C#
- C++

Frameworks:

- FastApi • Nest.js • ASP.NET Core • Temporal
- React.js • Express.js

Databases:

- PostgreSQL • DynamoDB

Familiar Technologies:

- GraphQL • Capacitor.js
- Docker • Prisma

OPEN SOURCE

- **SButtons**: Designed a 'Disco' button using **Less/CSS**, featured in their *special category*.
- **Kiwix**: Migrated **Java** to **Kotlin** for the Kiwix Android app, enhancing modernization and efficiency. *Pull Request*

EXPERIENCE

MCKINSEY & COMPANY | SOFTWARE ENGINEER I

Jan 2024 - Present | Bengaluru

- Designed and built an **LLM-powered claims verification pipeline** for automated **MLR review**, orchestrating **10+ dependent stages** (extraction, matching, relevance scoring, dedup, fact-checking, reference validation) as a **fault-tolerant Temporal workflow** that replaced manual claim-by-claim reviewer triage and cut review turnaround by **~60-70%**.
- Built the chat pipeline for a **Rep Copilot** serving **800+ pharma field reps**; implemented **LLM-based intent classification** with few-shot prompting to route requests across **Temporal workflows** with **450ms avg orchestration latency**; engineered a custom **polling mechanism** delivering streaming-like UX with **3.2s TTFT**; configured **KEDA autoscaling** on activity queues achieving **100% success rate** across load tests.
- Developed an **AWS IoT Core MQTT** service to provision and register patient sensors across **10+ hospital sites**; implemented **mutual TLS auth** for device identity verification and **manual QoS acknowledgment** to guarantee **zero-message-loss** during sensor onboarding, persisting device metadata to **PostgreSQL**.
- Built an **AWS IoT Core** message publishing service that routes **EventBridge** events to **IoT edge devices** over **MQTT**; implemented **multi-tenant topic addressing** across **10+ hospital sites** with **fault-tolerant retry handling**.
- Engineered a **reusable EventBridge publisher** abstraction as the central event bus integration layer for a real-time patient monitoring platform; designed a **single reusable publish function** adopted by **3-4 engineers** to onboard new clinical event types with **zero changes** to core infrastructure, supporting **15 clinical event types**.

DEVTRON | SOFTWARE DEVELOPMENT ENGINEER 1

Aug 2023 - Jan 2024 | Gurugram

- Engineered responsive, user-centric UI components and refactored legacy code for a **CI/CD dashboard** serving **15+ clients**, cutting system errors by **30%** and resolving dependency issues to improve overall software stability and reliability.

PROJECTS (HYPERLINKED)

LIFTLAB | REACT.JS, CAPACITOR.JS, SUPABASE, WEB/IOS/ANDROID

your training, modeled and queryable.

- Modeled **unified training load** across lifting and running instead of raw volume: per-set effort from **Epley 1RM** and **reps-in-reserve**, run load weighted by activity type/duration, combined into **ATL/CTL/TSB/ACWR** via **exponentially-weighted rolling averages** (**Banister-style** model used in pro training tools).
- Built an **LLM training assistant** with the full app state as context, including every gym session, split, and run, so it can answer synthesis questions a single log entry can't ("has my squat plateaued," "how does mileage affect my lifts") instead of just retrieving facts.
- Built a **Python evaluation harness** (**DeepEval**, **Gemini-as-judge**) to hold the chatbot to a bar beyond "looks fine": custom **G-Eval rubrics** scoring numeric-data faithfulness and whether an answer required real cross-session synthesis vs. trivial lookup, run against the live edge function (not a reimplement) to curate a benchmark of high-value questions.
- Grounded and iterated the load model against my own training log (**30+ real sessions**) rather than synthetic data, tweaking constants against real exercise-science literature and whether the output matched actual felt effort, not a spec.